

KOPANO LABS x CARS4MARS AFRICAN ROVER CHALLENGE 2026

DFR-01 PUBLIC WEB COPY

Submitted baseline: 02 August 2026

CURRENT-STATE DECLARATION

DFR-01 is design-locked. Procurement, fabrication, integration and physical testing are not complete. Planned components and diagrams are not test evidence.

REQUIREMENTS STATE

Drive: NO - no assembled rover exists.

Turn: NO - skid-steer architecture designed, not physically validated.

Battery operation: NO - 24 V 20 Ah LiFePO4 system selected, not procured/run.

Wireless control: NO - local Wi-Fi command path designed; range testing pending.

Live video: NO - RealSense D455 to Jetson RTSP/GStreamer path designed.

Payload transport: NO - 1 kg central tray designed, not fabricated/tested.

DESIGN BASELINE

Six Rhino IG52 24 V gearmotors; six 250 mm wheels; passive rocker-bogie.

Target envelope: 700 x 650 x 500 mm. BOM target: 28 kg.

Jetson Orin Nano Super for perception; Teensy 4.1 retains motor authority.

RealSense D455 + RPLIDAR A2M12; Wi-Fi command/video; LoRa fail-stop heartbeat.

24 V 20 Ah LiFePO4, BMS, 60 A fuse, contactor and emergency stop.

CONTROL INVARIANT

No LLM or cloud service is in the motor actuation loop.

KOPANO LABS x CARS4MARS - DFR-01 PUBLIC WEB COPY

OPEN BUILD STATE

Designed: LOCKED - DFR-01 submitted 02 Aug 2026.

Funded: PENDING EVIDENCE.

Ordered: NOT STARTED / NO ORDER EVIDENCE.

Assembled: NOT STARTED.

Tested: NOT STARTED.

NEXT VALIDATION GATES

Funding/orders -> frame, drive and protected power -> loaded drive/turn/E-stop.

Manual control + continuous onboard video -> one-take operator/rover evidence.

Payload, 30 cm bump, 45-degree ramp, range and object tests -> signed matrix.

Balloon sequence, autonomy, endurance and FMEA -> two sequences + 40-min run.

CONTROLLED R65,000 BUILD ALLOCATION

Drive/chassis R20,000 | Compute/control R11,000 | Vision R14,000

Power/safety R11,500 | Communications, spares and contingency R8,500

DISCLOSURE

Generative AI assisted source synthesis, language, software and labelled design visualisation. Engineering claims change only through team-controlled evidence.

This web copy is a concise public rendering of the source DFR-01 facts.

The authoritative six-page source report is preserved in the Kopano Labs archive.

Mission control: <https://kopanolabs.com/Cars4Mars/>